Food Webs and Network Ecology



Olari Eugenia, Goretti Francesca, Gonon Yoan, Katende Ukasha, Denis Drahokoupil

The Team



Francesca Goretti



Yoan Gonon



Denis Drahokoupil



Ukasha Katende



Eugenia Olari

Shah, P (2016) Convention on biological diversity and rural-urban connections with reference to Kenya











Vareschi, E. (1978). The ecology of Lake Nakuru

Primary Consumers in the Aquatic Food Web

Lesser Flamingo



Phoeniconaias minor

Most dominant consumer in biomass.

Specialized filter feeder consuming cyanobacteria (*Spirulina/Arthrospira*) in alkaline waters.

Soda Tilapia



Alcolapia grahami

Key alkaline-adapted cichlid species.

Introduced fish feeding directly on algae, essential to lake energy transfer.

Copepod



Lovenula africana

Major pelagic zooplankton.

Critical primary consumer driving planktonic biomass in the open water food web.

Chironomid Larvae



Leptochironomus deribae

Abundant benthic insect larvae.

Consumes organic sediment and algae, supporting benthic and aquatic food chains.

Rotifer



Brachionus dimidiatus

High consumption rate grazer.

Though small in individual biomass, possesses extremely high metabolic and turnover rates.

Vareschi, E. (1978). The ecology of Lake Nakuru

Secondary Consumers in the Aquatic Food Web

Great White Pelican



Pelecanus onocrotalus

Specialized piscivore driving energy transfer from fish populations to higher avian trophic levels.

Feeds predominantly on introduced Soda Tilapia (Alcolapia grahami) in open shallow waters.

Greater flamingo



Phoenicopterus roseus

Generalist filter feeder inhabiting alkaline lake shorelines and shallow waters.

Consumes a broader opportunistic diet including small invertebrates, aquatic insect larvae, and micro-algae.

African Fish Eagle



Haliaeetus vocifer

Apex avian predator positioned at the top of the aquatic food chain interface.

Hunts surface-swimming fish and waterbirds from high acacia tree perches surrounding the lake shore.

Vareschi, E. (1978). The ecology of Lake Nakuru

Tertiary Consumers in the Aquatic Food Web

BIRDS OF PREY (GREIFVÖGEL)



Explicitly cited in literature as ***Konsumenten höherer Ordnung*** (higher-level consumers).

Apex avian predators that prey on fish-eating waterbirds and other terrestrial or aquatic vertebrates.

MARABOU STORKS (MARABUS)



Leptoptilos crumenifer

Large opportunistic scavengers and carnivores occupying top trophic positions, feeding on fish, waterbird nestlings, and carrion.

"Regarding tertiary consumers, ecological literature is relatively vague and typically refers to them broadly as 'higher-level consumers.' Due to the short food chain in Lake Nakuru, top predators consist primarily of Birds of Prey and Marabou Storks preying on fish-eating birds or other vertebrates."

Vareschi, E. (1978). The ecology of Lake Nakuru

Terrestrial Trophic Dynamics & Lake Connections

PRIMARY CONSUMERS / HERBIVORES



The **Giraffe** (Giraffa camelopardalis), **Zebra** (Equus quagga), **African Buffalo** (Syncerus caffer), and **Black Rhinoceros** (Diceros bicornis) feed on surrounding vegetation rather than lake production.

PREDATORS



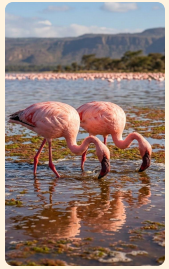
Apex predators such as the **Lion** (Panthera leo), **Leopard** (Panthera pardus), and **African Rock Python** (Python sebae) influence herbivore distribution.

"Terrestrial herbivores feed on vegetation, while predators feed on other animals. These terrestrial food webs are connected to the lake ecosystem through nutrient movement and animal activity."

Vareschi, E. (1978). The ecology of Lake Nakuru

Trophic Structure & Aquatic Ecosystem Roles

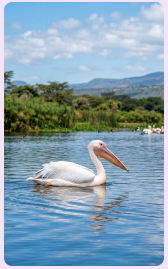
Primary Consumers



Lesser Flamingo (*Phoeniconaias minor*) & Soda Tilapia (*Oreochromis grahami*).

Feed directly on primary producers in alkaline aquatic environment.

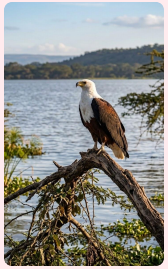
Secondary Consumers



Great White Pelican (*Pelecanus onocrotalus*) & Great Cormorant (*Phalacrocorax carbo*).

Act primarily as specialized piscivores within food web.

Tertiary Consumers



Trophic level dynamically depends on target prey availability.

Highlights feeding role flexibility and adaptive predation strategies across ecosystem structure.

Decomposers



Microscopic bacteria & fungi.

Essential for recycling organic nutrients back into the ecosystem from dead organic matter.

Producers



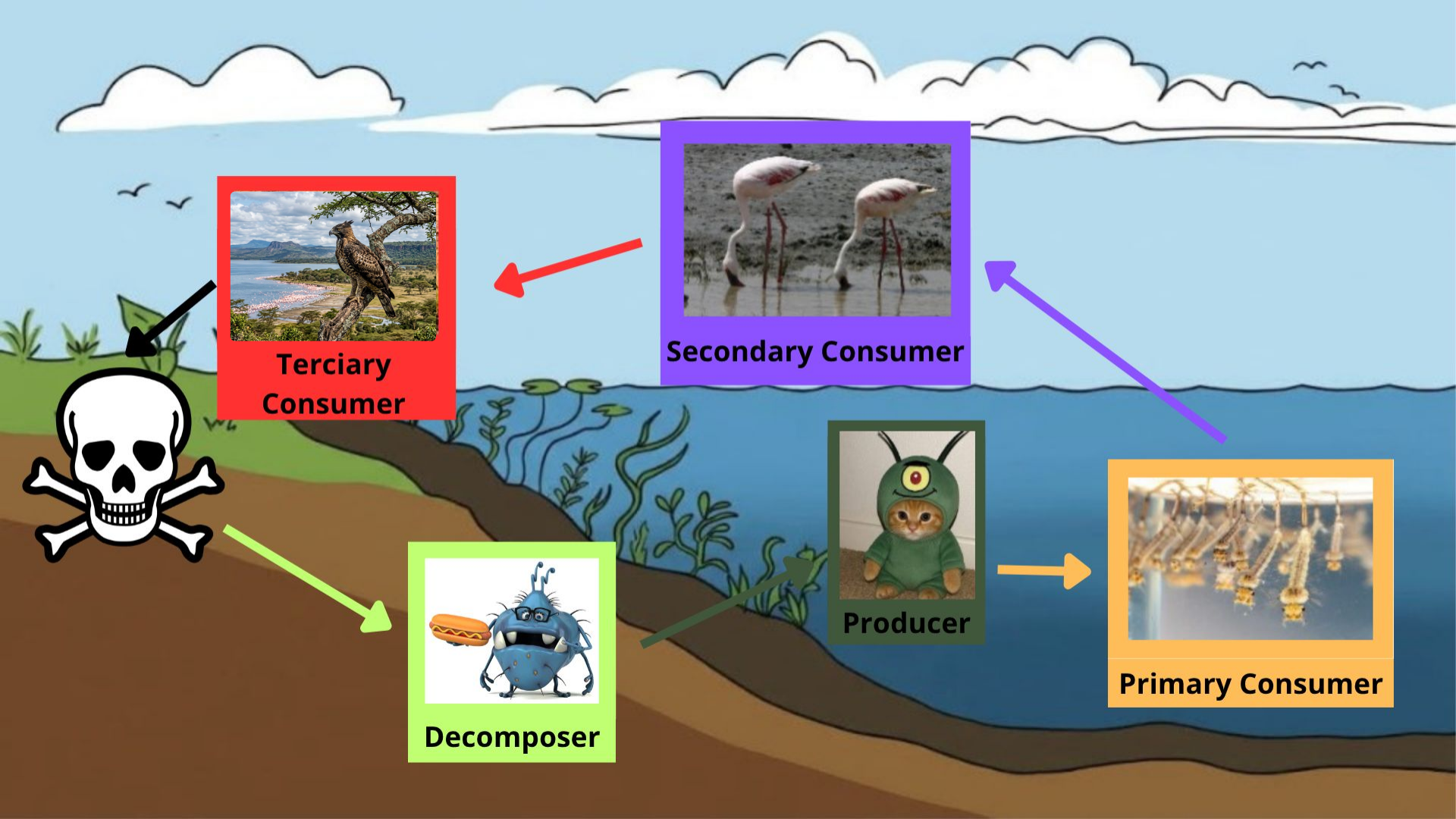
Microscopic algae & cyanobacteria, primarily *Spirulina platensis* (formerly *Arthrospira*).

Thrive abundantly in highly alkaline waters.

What is a Food Chain?

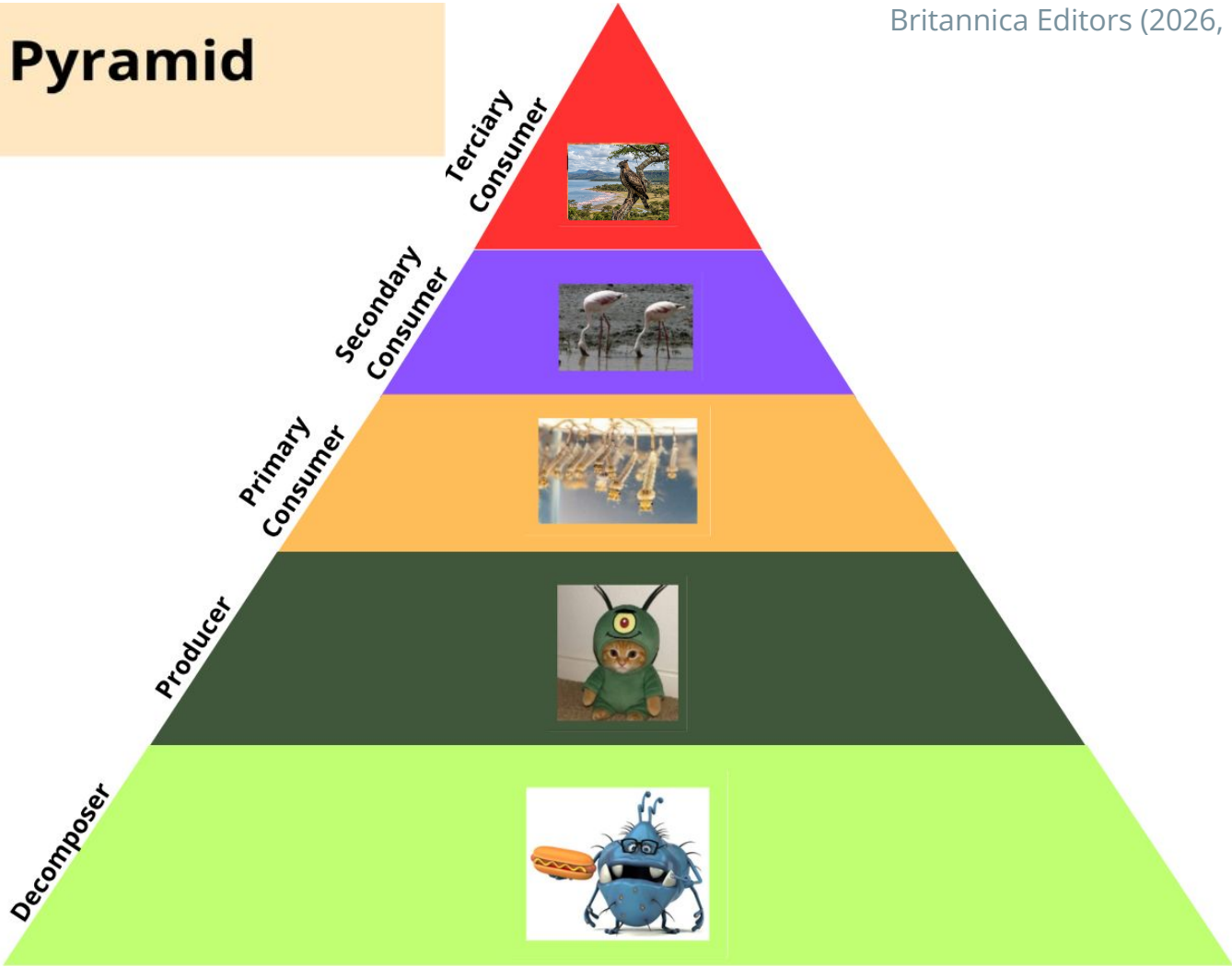
- Graphical representation of nutrients moving in the Ecosystem
- Who eats whom
- Where nutrients start and end up




Terciary Consumer
Secondary Consumer
Producer
Primary Consumer
Decomposer

Trophic Pyramid

Britannica Editors (2026, September 2).

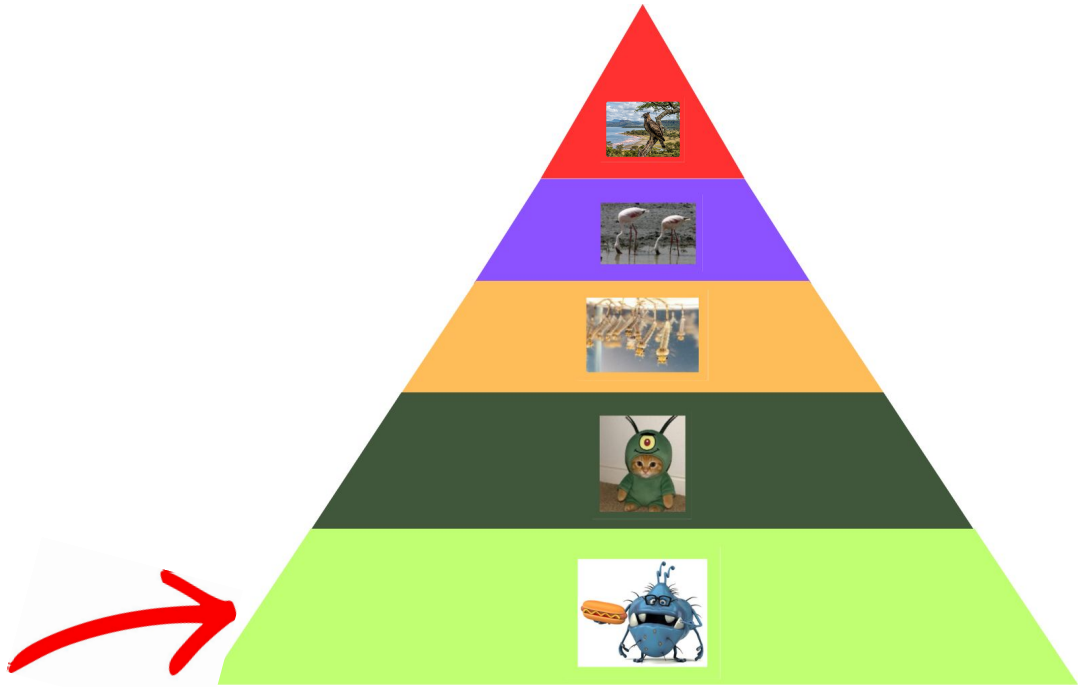


Sottosanti, K. (2026, October 10).

Decomposers

- Fungi/Bacteria
- Breakdown of dead organisms
- Provide nutrients to producers
- Bedrock of the food chain

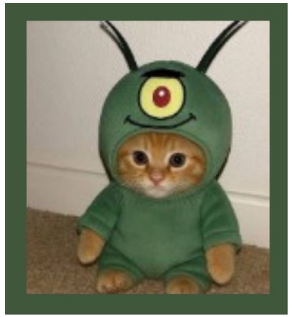


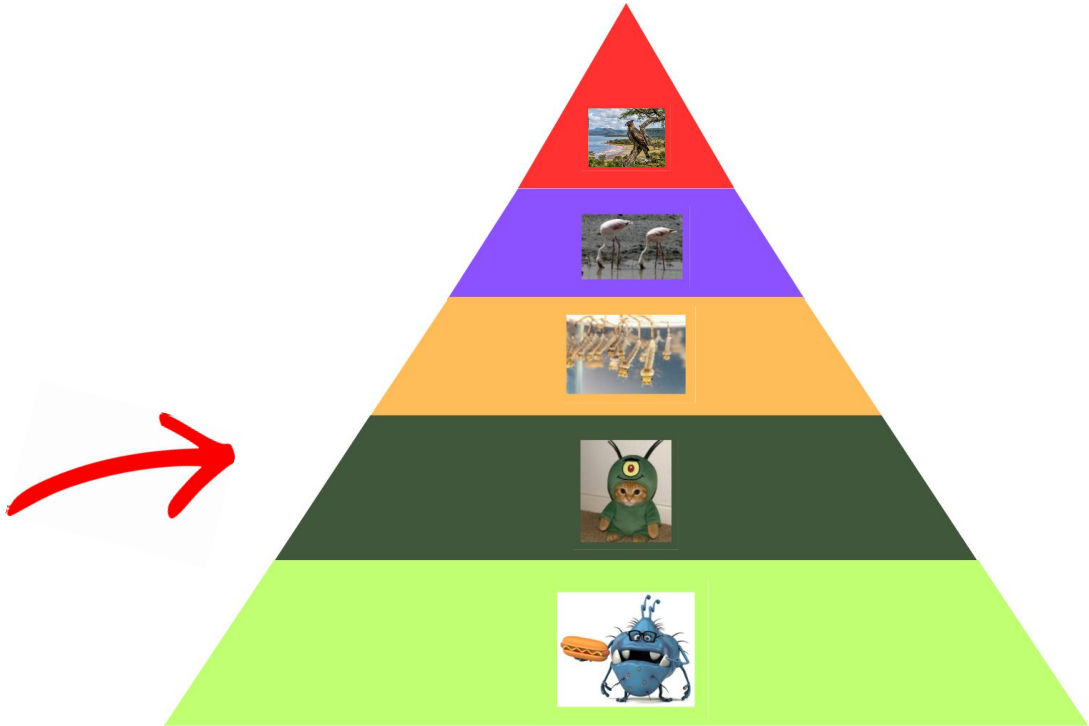


Sottosanti, K. (2026, October 10).

Producers

- Plants/algae
- Turn Energy from the Sun to nutrients
- Autotrophs
- Base of the food chain



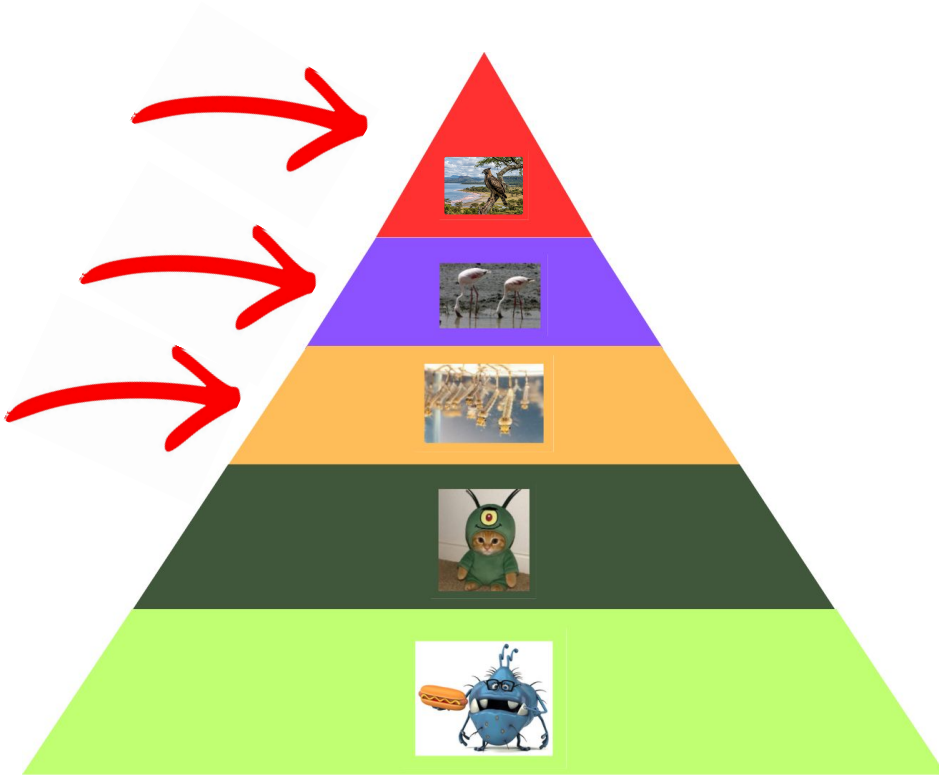


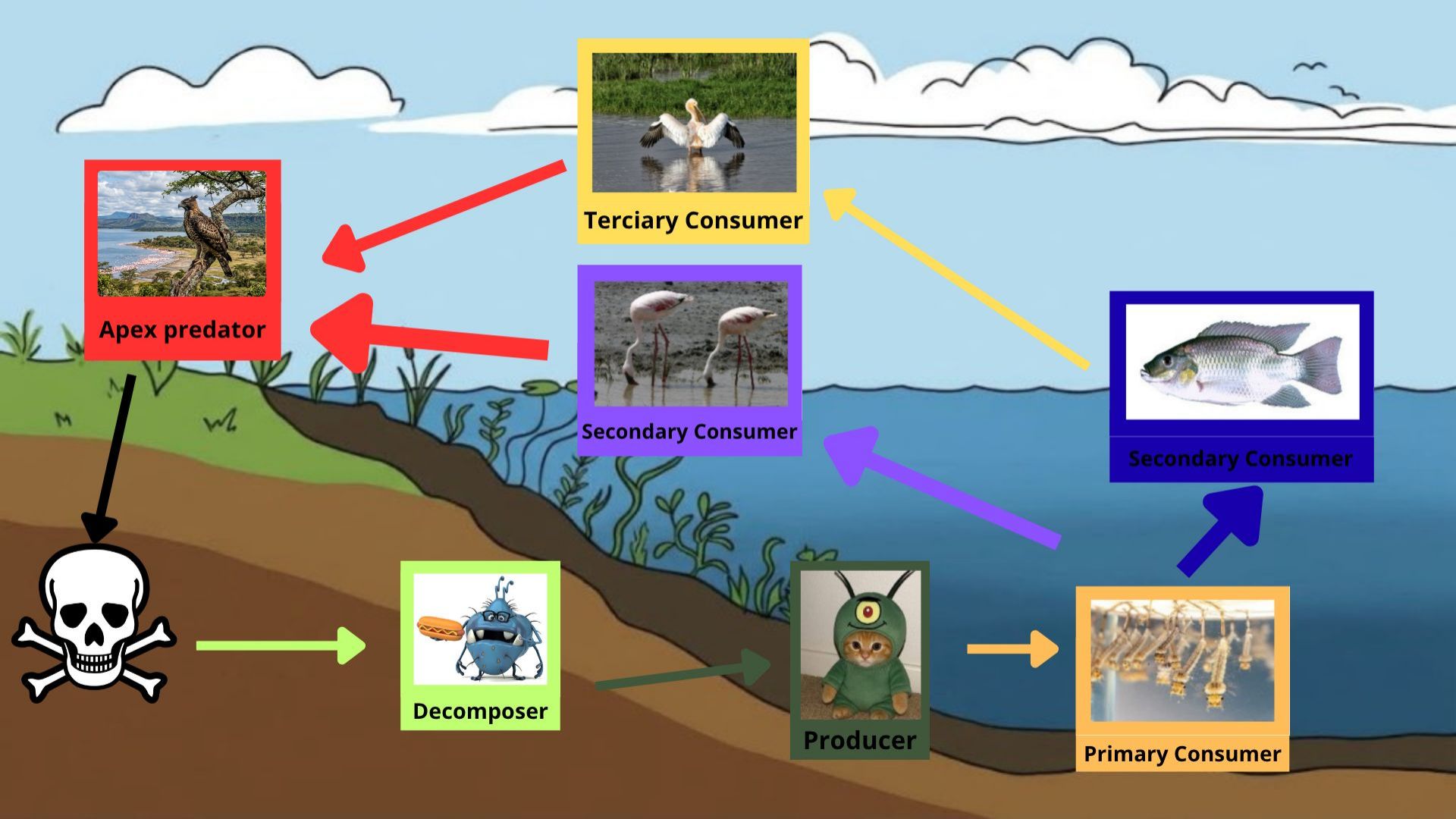
Sottosanti, K. (2026, October 10).

Consumers

- Herbivores/predators
- Trophic levels
- Get energy from producers or fellow consumers
- Heterotrophs

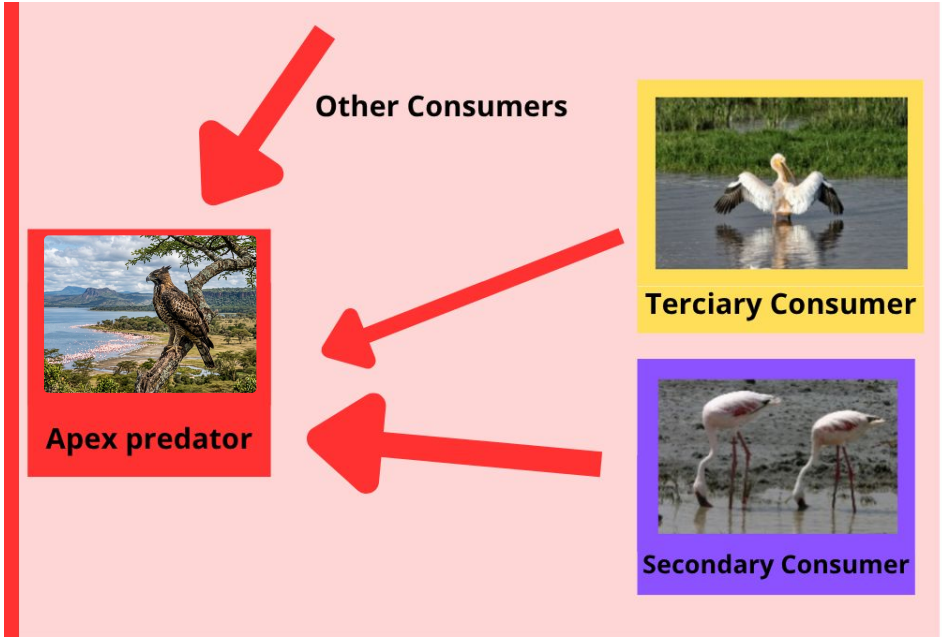




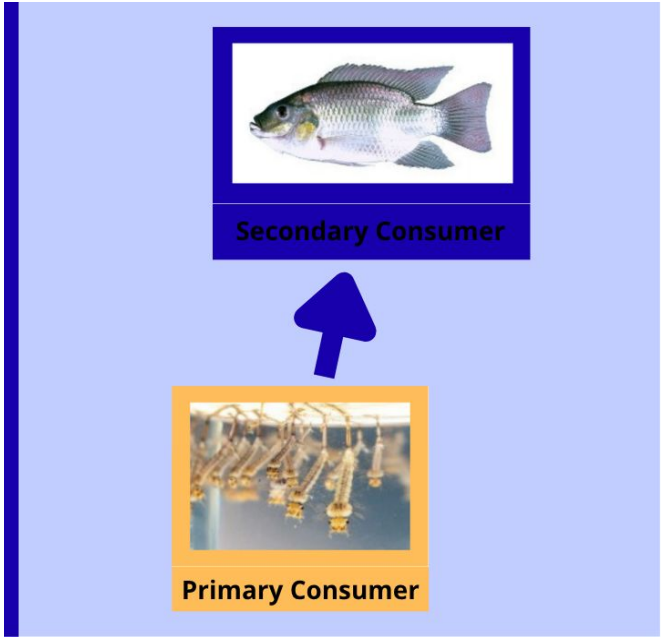

Terciary Consumer
Apex predator
Secondary Consumer
Secondary Consumer
Decomposer
Producer
Primary Consumer

Feeding relationships

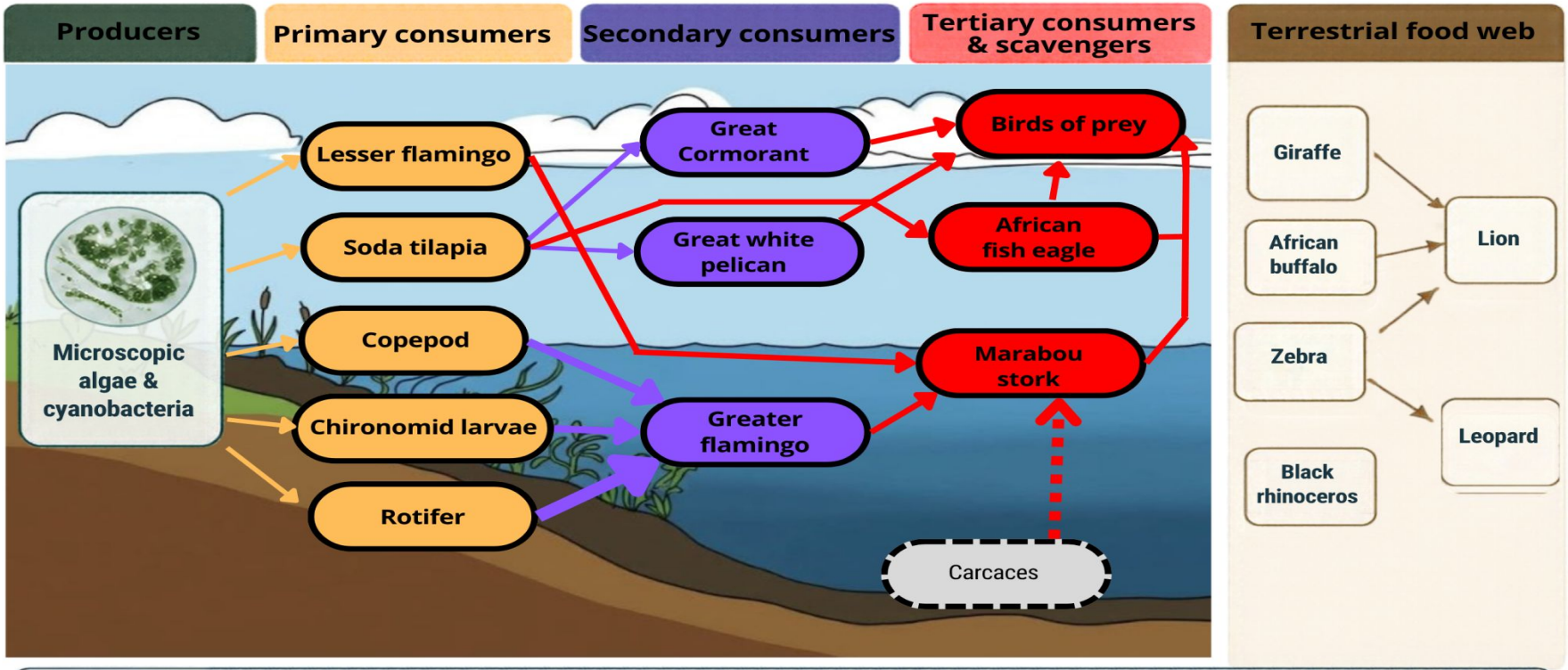
Weak



Strong



Lake Nakuru Food Web



KEYSTONE SPECIES

- Small biomass
- Maintains balance of the food web
- Role-based impact (domino effect)

Thompson, J.N. (2023, October 9).

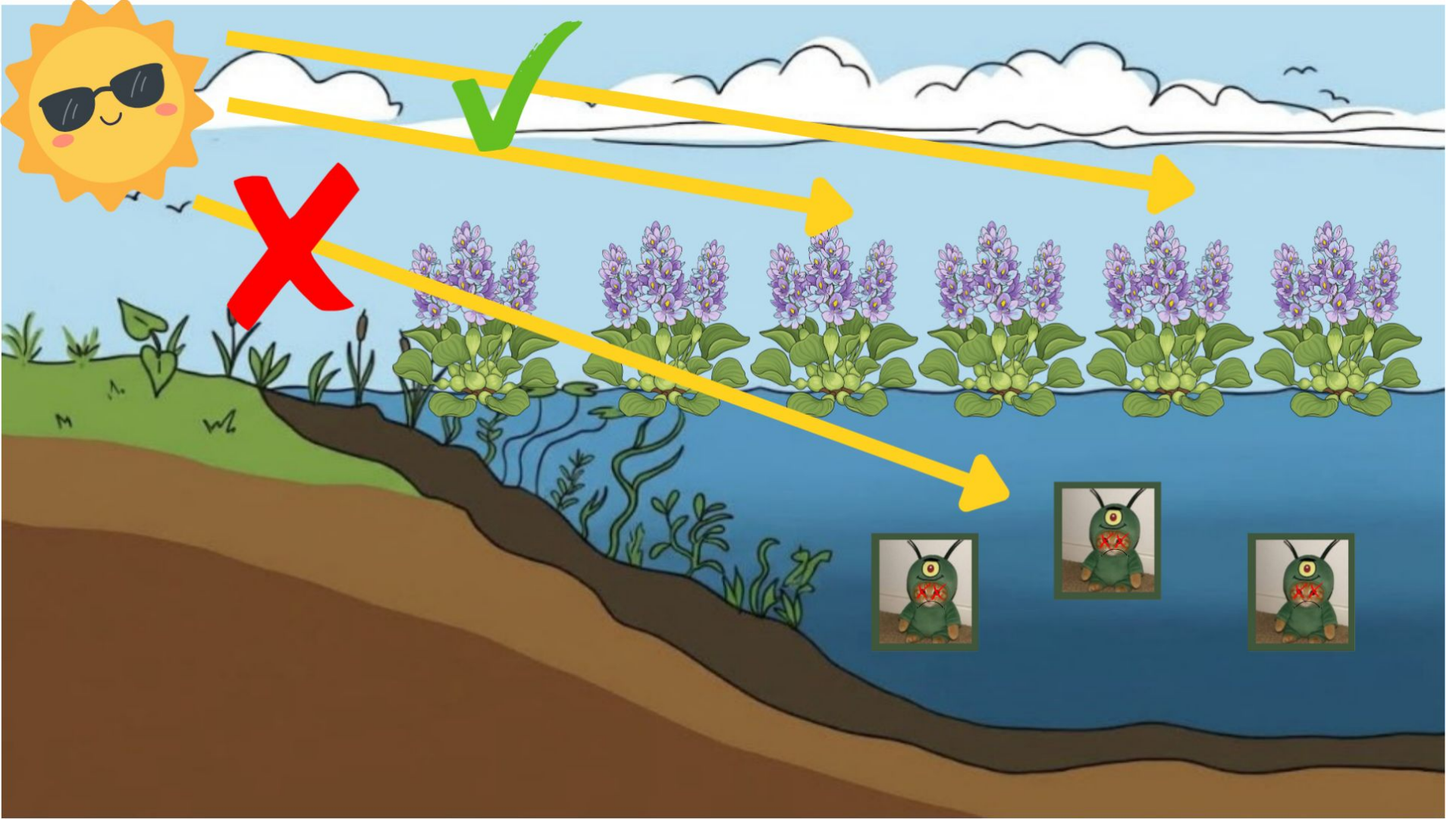


DOMINANT SPECIES

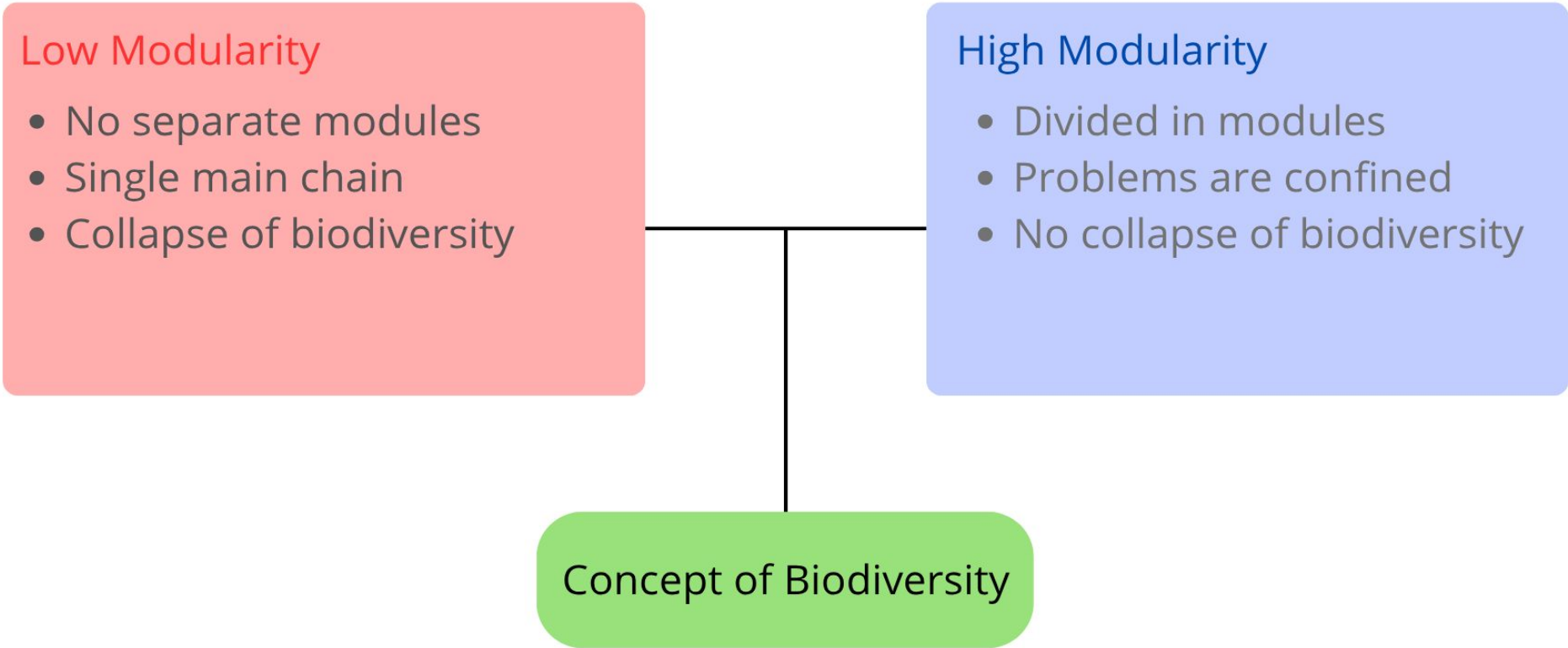
- Very high biomass / abundance
- Control ecosystem energy
- Impact based on quantity

Smee, D. (2010)



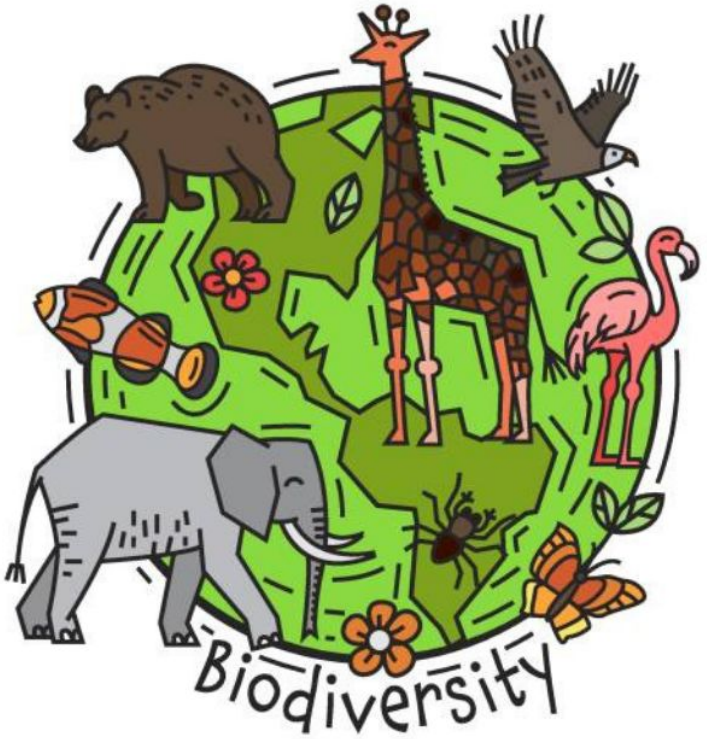


Network Modularity



What is Biodiversity ?

- The variety of life in Ecosystems
- Quantified by the number of species or genes

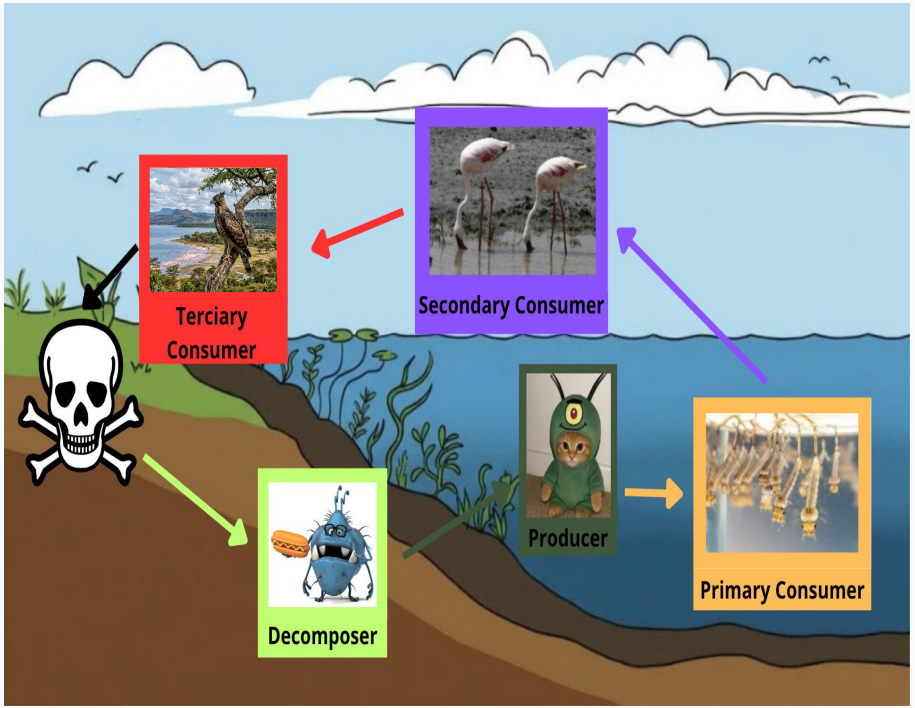


NETWORK RESILIENCE

Food Web Complexity & Biodiversity

Paine, R. T. (1966)

K.R.S. *et al.* (2026)



Multiple Feeding Links

Diverse species relationships allow greater ecological coexistence.

Alternative Prey Routes

Diet switching maintains energy flow during single-source declines.

Ecological Niche Variety

Varied resource usage buffers ecosystems against direct competition.

Complexity supports resilience, though highly complex webs are not automatically stable.

How Does Food Web Complexity Affect Biodiversity ?

Vol. 100, No. 910 The American Naturalist January–February, 1966

FOOD WEB COMPLEXITY AND SPECIES DIVERSITY

ROBERT T. PAINE

Department of Zoology, University of Washington, Seattle, Washington

Though longitudinal or latitudinal gradients in species diversity tend to be well described in a zoogeographic sense, they also are poorly understood phenomena of major ecological interest. Their importance lies in the derived implication that biological processes may be fundamentally different in the tropics, typically the pinnacle of most gradients, than in temperate or arctic regions. The various hypotheses attempting to explain gradients have recently been reviewed by Fischer (1960), Simpson (1964), and Connell and Orias (1964), the latter authors additionally proposing a model which can account for the production and regulation of diversity in ecological systems. Understanding of the phenomenon suffers from both a specific lack of synecological data applied to particular, local situations and from the difficulty of inferring the underlying mechanism(s) solely from descriptions and comparisons of faunas on a zoogeographic scale. The positions taken in this paper are that an ultimate understanding of the un-

More complex equals better ?



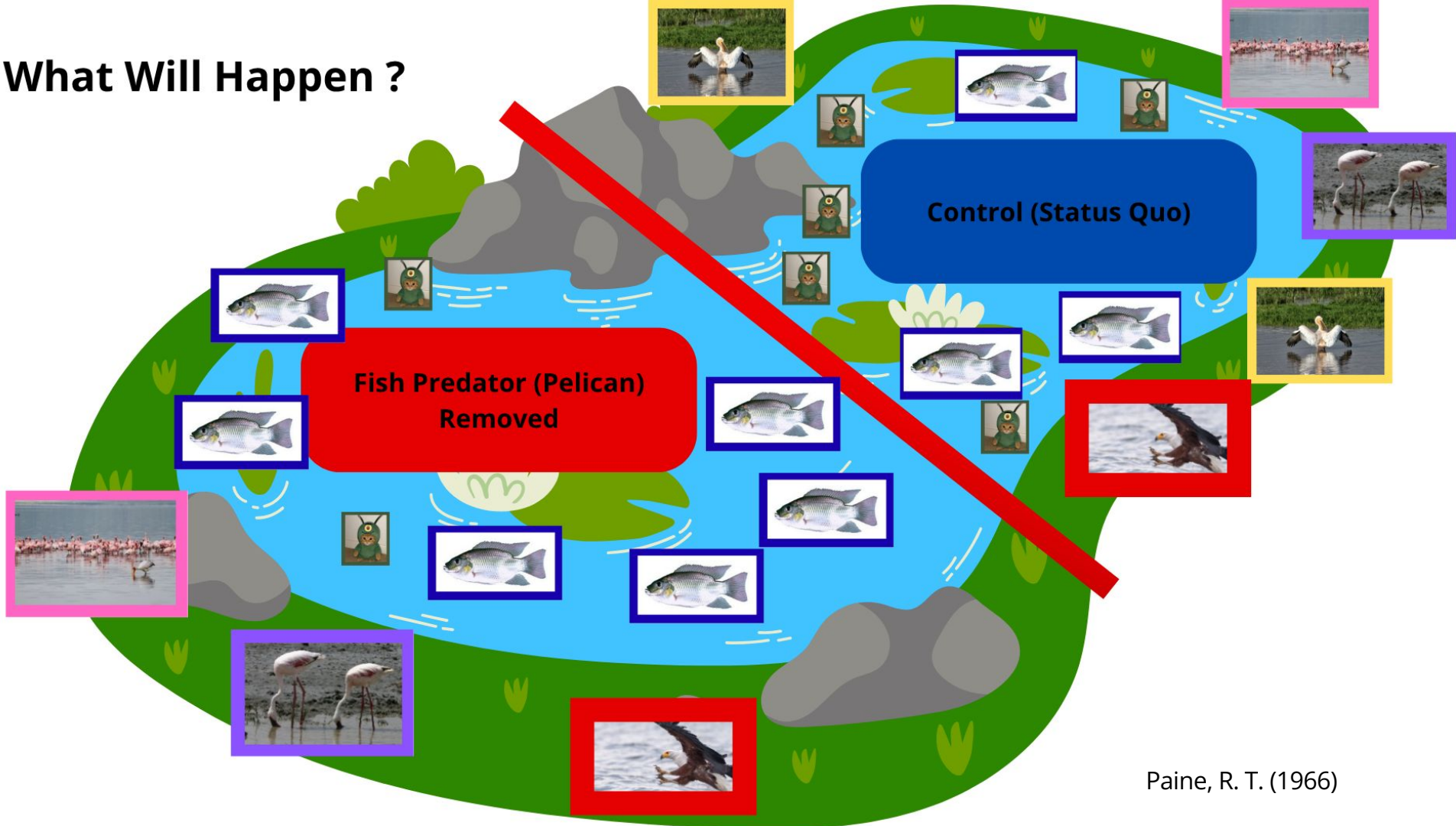


What if complexity changes ?

Less complex equals better ?

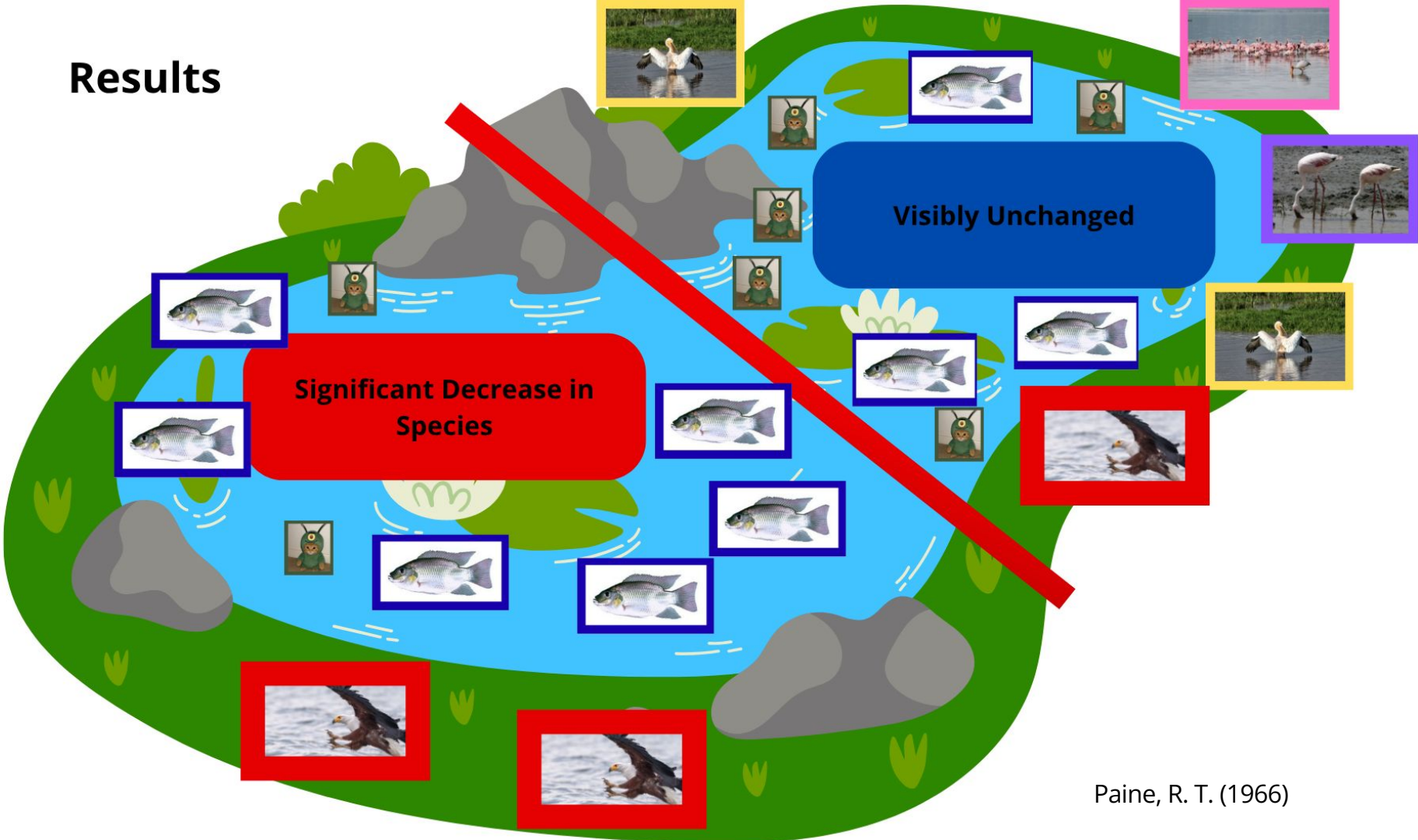


What Will Happen ?



Paine, R. T. (1966)

Results



Paine, R. T. (1966)

Summary of what happened

- Key predators = they make sure no species take over
- Predators and biodiversity (indirect impacts = predators don't just eat them but they make sure that species can have space to live and grow)
- Predators necessarily don't have the same effect
- Biomass vs biodiversity (even if biomass increases = fish, biodiversity decreases = flamingo)
- The amount of energy can set limits on diversity

Paine, R. T. (1966)

What is Rewiring ?

- Reorganization of how energy moves in the ecosystem

- 2 types of rewiring

Topological

- The actual feeding connections change
- Conections can be added or removed

Interaction-strength

- The actual feeding connections remain the same
- Strength of connection changes

Bartley-et-al.-2019

What can cause rewiring ?

Bartley-et-al.-2019

Change in food availability

- Sources of energy change in availability
- Animals are forced to change the strenght of connections

Invasive species

- A new species is introduced
- Forces ecosystem to create new or get rid of connections

Changed habitat conditions

- Warming, drying, flooding....
- Animals have to relocate changing their connections

Phenological mismatch

- Timming of energy sources changes
- Animals dependant on timing are forced to change

ECOSYSTEM DISRUPTION

How Invasive Species Rewire Food Webs

Resource Competition
Native species lose food access as invaders establish rapid dominion.

Direct Predation
Vulnerable local populations undergo steep declines due to new pressure.

Indirect Effects
Cascade changes affect organisms far removed from direct interaction.

Network Disruption
Interconnected web links destabilize across multiple trophic levels.

The Team



Francesca Goretti

-Food web
-Dominant and keystone species
- Network modularity



Yoan Gonon

- Introduction of the ecosystem



Denis Drahokoupil

- Food Chains
- Biodiversity
Example



Ukasha Katende

- Rewiring



Eugenia Olari

Introduction of the ecosystem

Questions ?

Sources Library

Vareschi, E. (1978). The ecology of Lake Nakuru (Kenya). I. Abundance and feeding of the lesser flamingo. Oecologia, 32(1), 11–35. https://doi.org/10.1007/BF00344687

Vareschi, E., & Vareschi, A. (1984). The ecology of Lake Nakuru (Kenya). IV. Biomass and distribution of consumer organisms. Oecologia, 61(1), 70–82. https://doi.org/10.1007/BF00379091

Vareschi, E., & Jacobs, J. (1984). The ecology of Lake Nakuru (Kenya). V. Production and consumption of consumer organisms. Oecologia, 61(1), 83–98. https://doi.org/10.1007/BF00379092

Vareschi, E., & Jacobs, J. (1985). The ecology of Lake Nakuru. VI. Synopsis of production and energy flow. Oecologia, 65(3), 412–424. https://doi.org/10.1007/BF00378917

Kaggwa, M. N., Gruber, M., Oduor, S. O., & Schagerl, M. (2013). A detailed time series assessment of the diet of lesser flamingos: Further explanation for their itinerant behaviour. Hydrobiologia, 710, 83–93. https://doi.org/10.1007/s10750-012-1105-1

Charlotte A. Ward, Tyler D. Tunney, Kayla R. S. Hale, Reilly F. O'Connor & Kevin S. McCann: The rewiring of ecological networks in a variable world https://www.nature.com/articles/s44358-026-00159-9

Kerlley Bong Siew Sin: Dissappearing pink flamingos and evolution of lake nakuru https://sunwayuniversity.edu.my/explore/thinkpieces/disappearing-pink-flamingos-and-evolution-of-lake-nakuru

Paine, R. T. (1966). Food Web Complexity and Species Diversity. *The American Naturalist*, *100*(910), 65–75. http://www.jstor.org/stable/2459379

Sottosanti, K. (2026, October 10). food web. Encyclopedia Britannica. https://www.britannica.com/science/food-web

Britannica Editors (2026, September 2). trophic pyramid. Encyclopedia Britannica. https://www.britannica.com/science/trophic-pyramid

Thompson, J.N. (2023, October 9). keystone species. Encyclopedia Britannica. https://www.britannica.com/science/keystone-species

Smee, D. (2010) Species with a Large Impact on Community Structure. *Nature Education Knowledge* 3(10):40

Bartley, T.J., McCann, K.S., Bieg, C. *et al.* Food web rewiring in a changing world. *Nat Ecol Evol* **3**, 345–354 (2019). https://doi.org/10.1038/s41559-018-0772-3

Images:

- https://www.wildlifeofkenya.com/categories/wading-and-water-birds/pelican-great-white/
- https://www.inaturalist.org/places/lake-nakuru?__cf_chl_tk=FfSHRLRIm3DfN8k4XOtYz6XR3TszGcgTWHVAtYWaP2Y-1792263216-1.0.1.1-O1kZih3NaFUEDbeUODiqY5NdKjhOn7XF8PNEAIGfNIw#page=1&taxon=3
- https://sunwayuniversity.edu.my/explore/thinkpieces/disappearing-pink-flamingos-and-evolution-of-lake-nakuru